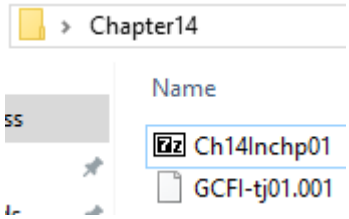
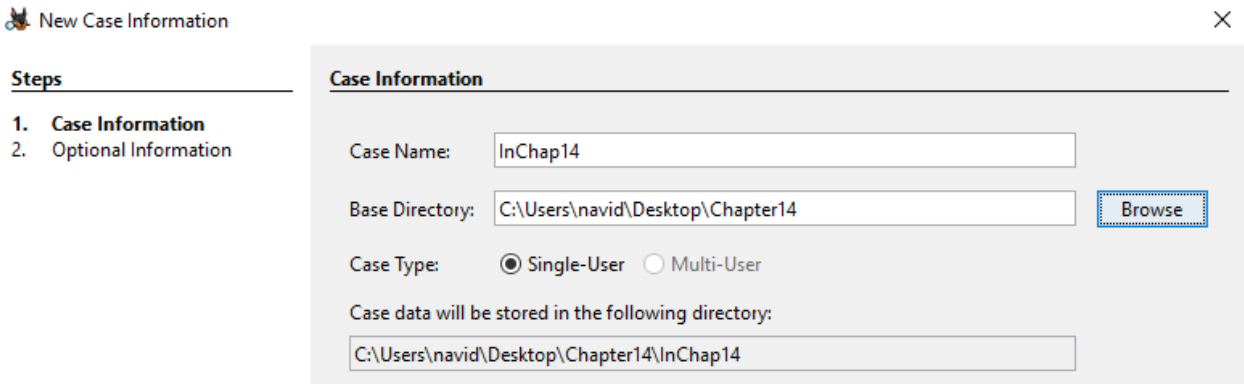
Navid Baghaei

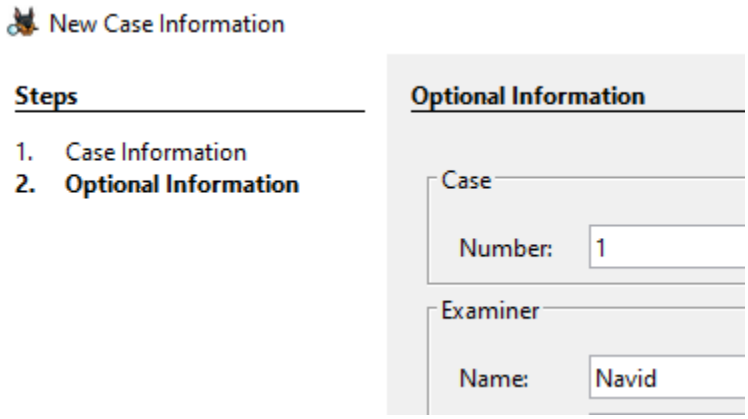
Lab 14 & 15

TASK 1

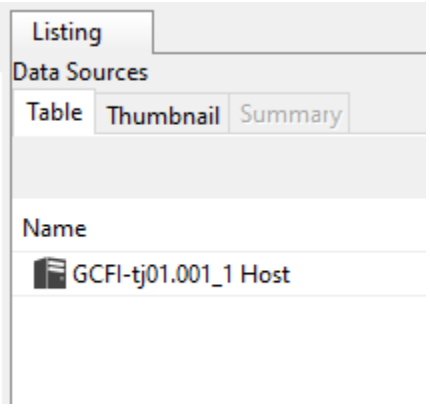
1. 

Extracted the image file.

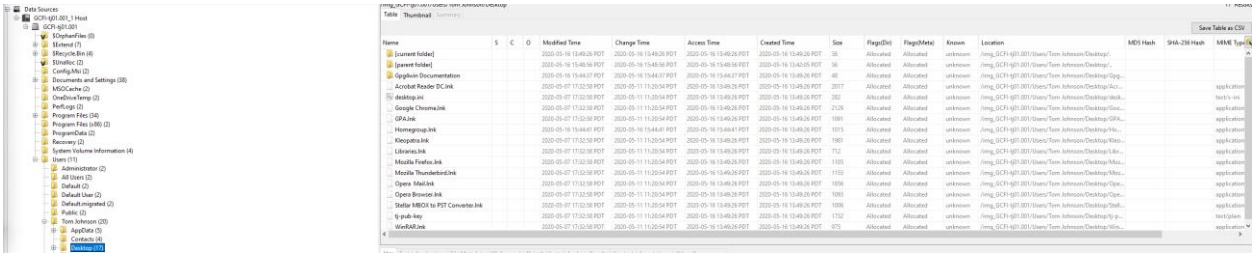




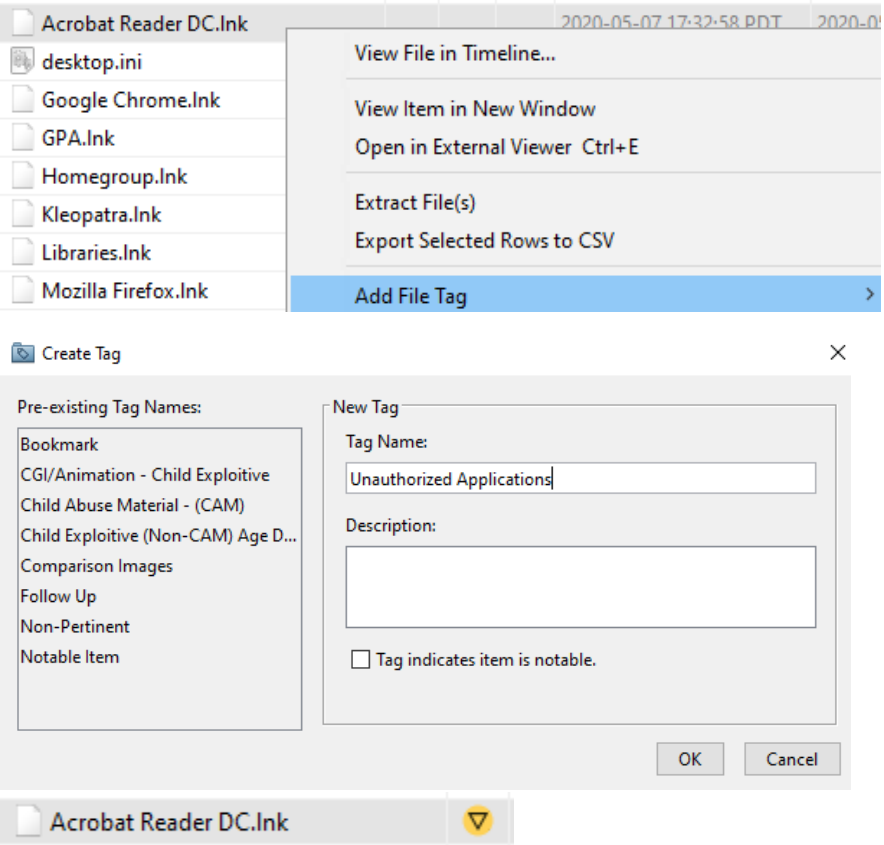
Created a case in autopsy named InChap14, with a case number of 1.



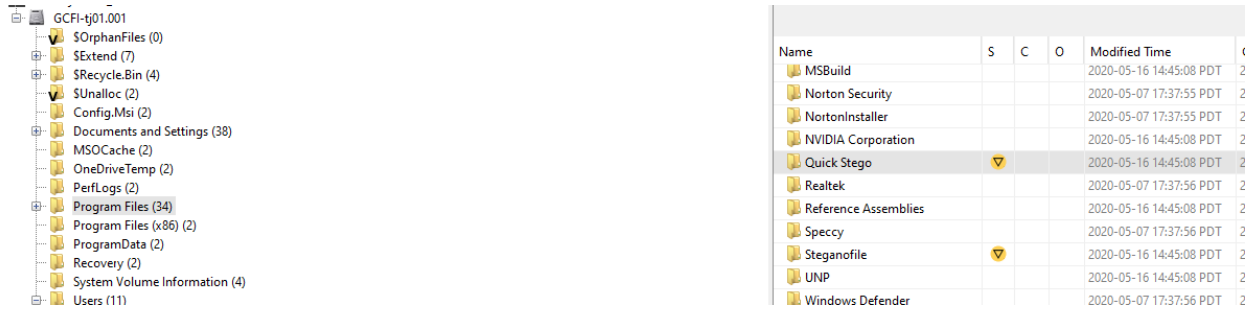
Added the “GCFI-tj01.001" Disk Image to the case data sources.

2. 

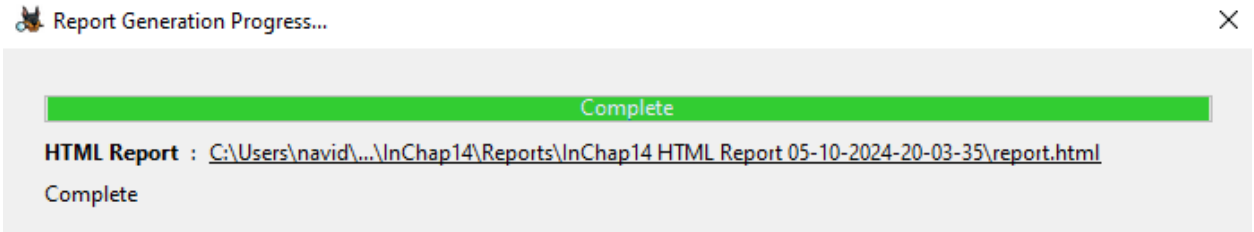
Expanded data sources until I reached Tom Johnson’s Desktop.



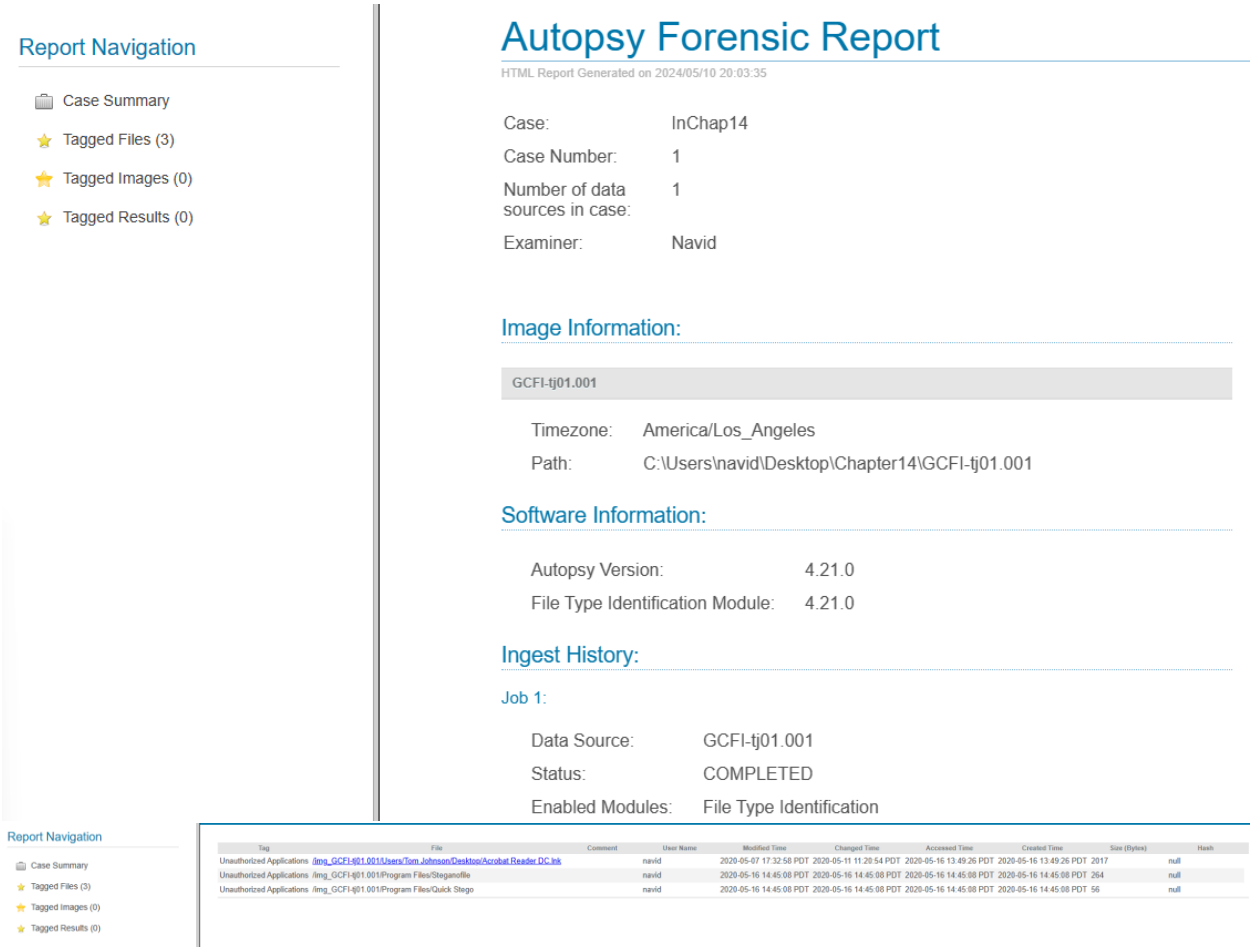
Added the tag “Unauthorized Applications” to Acrobat Reader DC.lnk.



Navigated to Program Files and added any steganography application folders to the Unauthorized Applications tag.

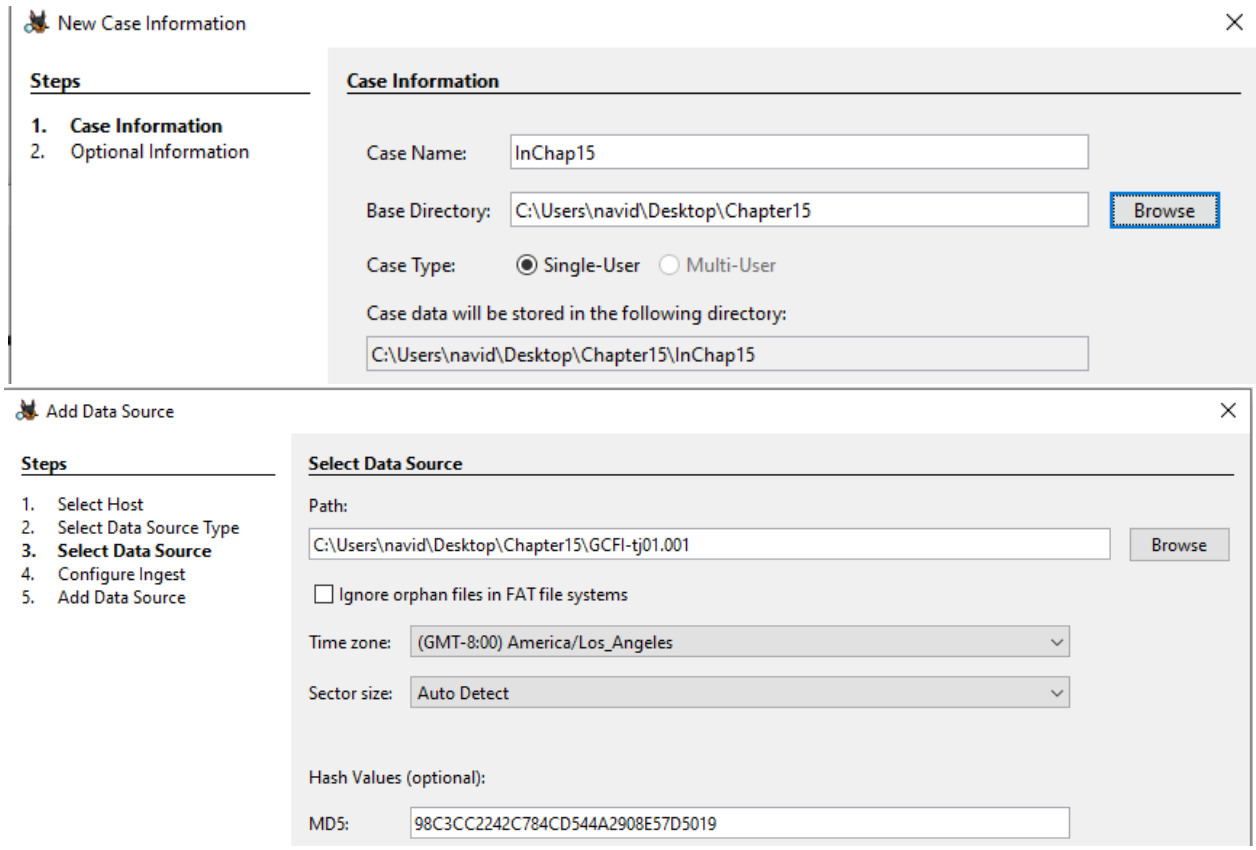
3. 

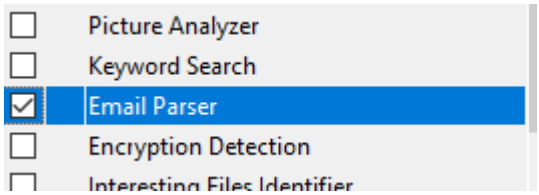
Created an HTML Report of all results.



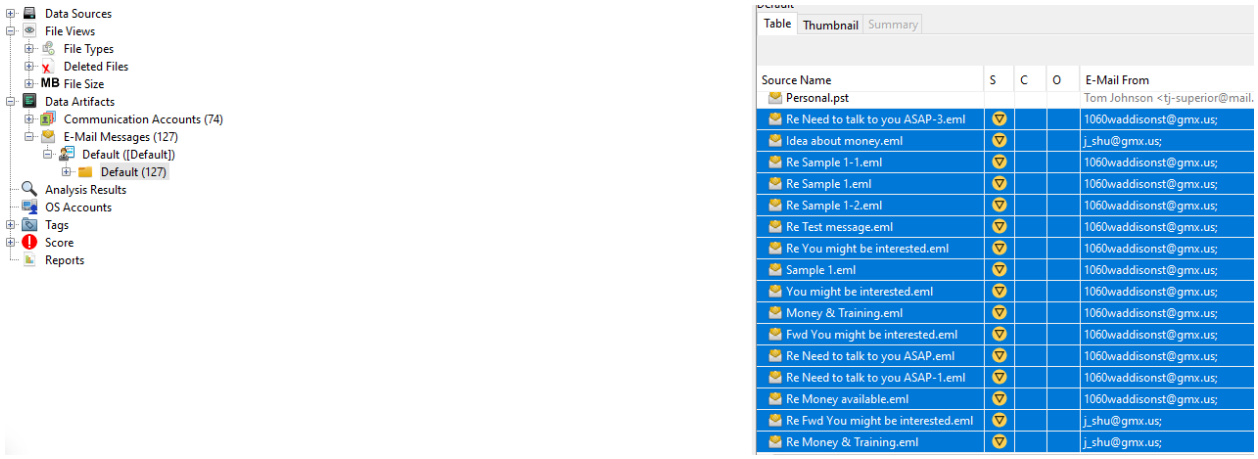
In the forensic examination of the system pertaining to case InChap14, important digital artifacts were discovered, notably several unauthorized applications designed for steganography, hiding information within other files. These applications were identified on the Desktop and in Program Files under the user profile 'Tom Johnson' and tagged as "Unauthorized Applications". The detection of these steganography applications is significant as they may be used to conceal illegal activities, breach data security, or take part in unauthorized communication. The evidence was collected using the Autopsy forensic tool, which enabled the identification and categorization of these suspicious files efficiently and transparently.

TASK 2

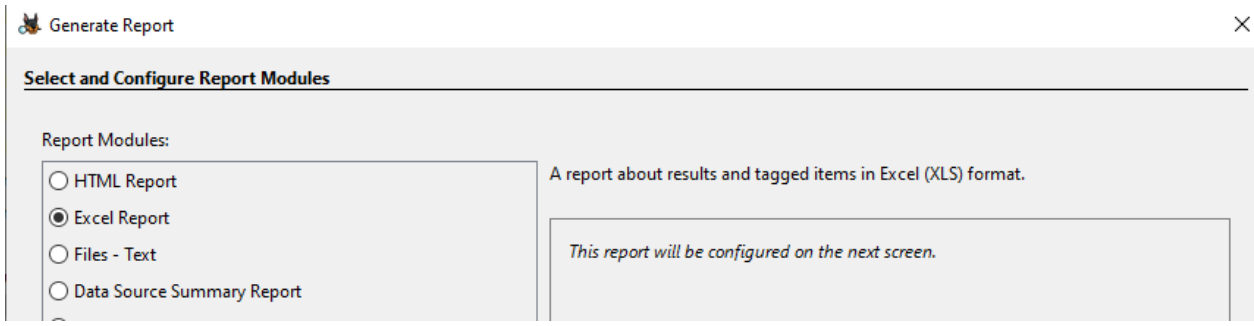
1. 



Created a new case “InChap15” and added the image file back with the hash “98C3CC2242C784CD544A2908E57D5019” for validation and chain of custody proof. Only selected the Email Parser.

2. 

Under the Data Artifacts, I opened the E-Mail messages and added the tag “GMX Email” to every email from someone with a gmx.us email address.

3. 

Generated an excel report of all the results.

Report:

1. The emails were extracted using a digital forensics process involving the Autopsy forensic tool. The tool was configured to specifically parse email data from the disk image of the suspect's computer. This included setting up the case with proper naming conventions and using the "Email Parser" ingest module to ensure all emails present within the disk image were captured accurately. The process was important to ensure that all relevant data was obtained without alteration, maintaining the integrity of the emails as they were stored on the disk.

2. The analysis of the extracted emails reveals ongoing negotiations over financial compensation and the secret handling of sensitive data. One email series clearly shows financial negotiations, with Tom Johnson expressing his concerns about the low compensation for the risks he's taking: "WHAT!! I'm risking my job for $100? Tell them I need more than that if they want this to go forward." This led to a revised offer of $500, as Jim Shu communicated: "Tom, I was able to get the investor up to $500, will this do?" The quick increase in the offered amount highlights the serious risks involved in the actions requested.

   Furthermore, another email provides details about illegal activities involving data manipulation. Tom Johnson describes a method to avoid email monitoring: "I smuggled these files out, they are JPEG files I edited with my hex editor so that the email monitor won't pick up on them. So, to view them you have to re-edit each file to the proper JPEG header of offset 0x FF D8 FF E0 and offset 6 of 4A." This email not only shows a deliberate breach of security protocols but also illustrates the techniques used to keep the operation secret.

   These examples from the emails not only illustrate potential unlawful activities and negotiations but also demonstrate the methods used to secretly manage and transmit sensitive information.

3. To make sure the digital evidence was not tampered with, we started by creating an MD5 hash value for the disk image as soon as we received it. This hash value works like a unique code that represents the original state of the data. Whenever we accessed or analyzed the data later, we checked it against this original hash code to confirm that nothing had changed. This method is a reliable way to track that the data remained the same from the time we got it until we finished our investigation. By consistently checking the evidence against the hash code, we made sure the information was secure and unchanged, keeping it trustworthy for legal use.